Logistic Regression

CIS 520: Machine Learning (Fall 2009)

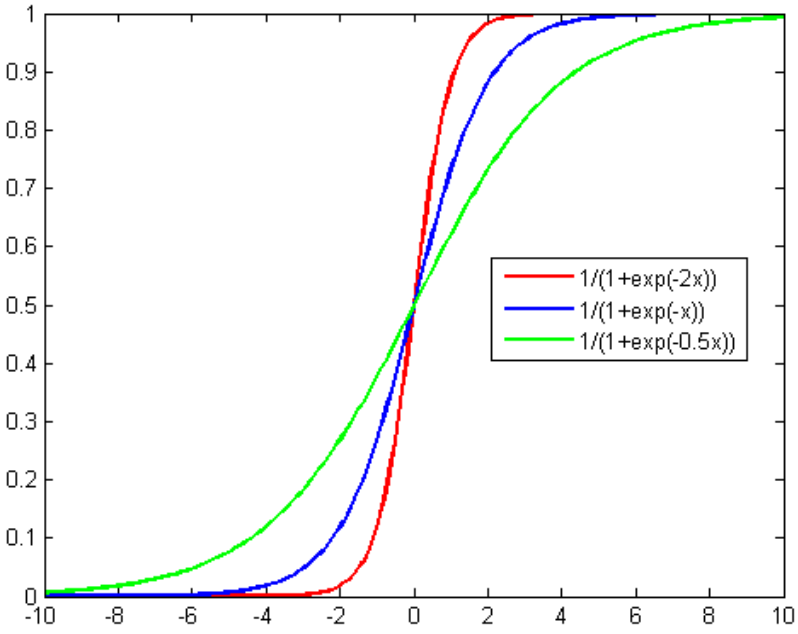
Archived lecture page

Source: https://web.archive.org/web/20100712095836/http://alliance.seas.upenn.edu/~cis520/wiki/index.php?n=Lectures.Logistic

We saw how a generative classifier, the Naive Bayes model, works. It assumes some functional form for $\hat{P}(X \mid Y)$ and $\hat{P}(Y)$ and estimates parameters of P from training data. It then uses Bayes rule to calculate $\hat{P}(Y \mid X = x)$. In generative models, computation of $P(Y \mid X)$ is always indirect, through Bayes rule. Discriminative classifiers, such as logistic regression, instead assume some functional form for $P(Y \mid X)$ and estimate its parameters from training data. We first consider binary classification, $y \in \{-1, 1\}$. The model uses the logistic function $f(x) = 1/(1 + e^{-x})$, which transforms a real-valued input into a value between 0 and 1:

$$P(Y = 1 \mid \mathbf{x}, \mathbf{w}) = \frac{1}{1 + \exp\{-\mathbf{w}^\top \mathbf{x}\}},$$
$$P(Y = -1 \mid \mathbf{x}, \mathbf{w}) = 1 - P(Y = 1 \mid \mathbf{x}, \mathbf{w}) = \frac{\exp\{-\mathbf{w}^\top \mathbf{x}\}}{1 + \exp\{-\mathbf{w}^\top \mathbf{x}\}}.$$

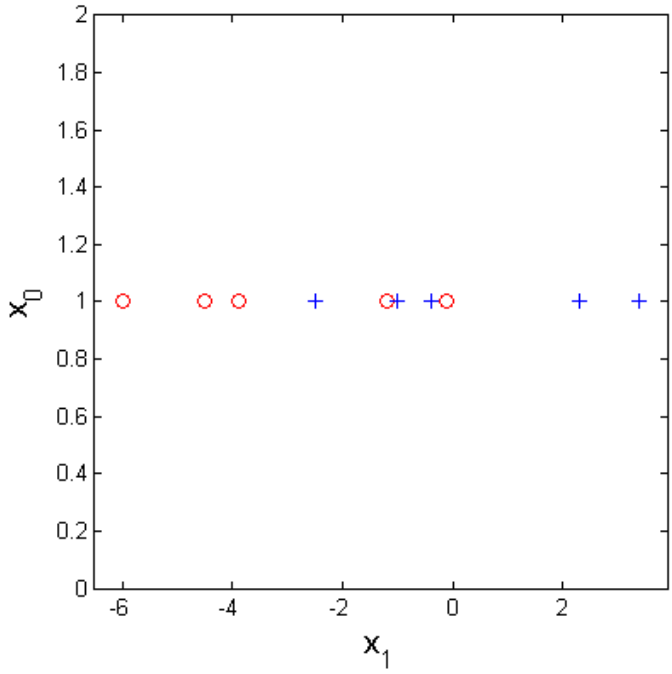
The logistic (or sigmoid) function is 0.5 when its argument is zero, and tends to 0 for negative infinity and 1 for positive infinity.



Logistic function.

1 1-D Example

Consider a 1-D problem, with positive examples marked as pluses and negative examples as circles.

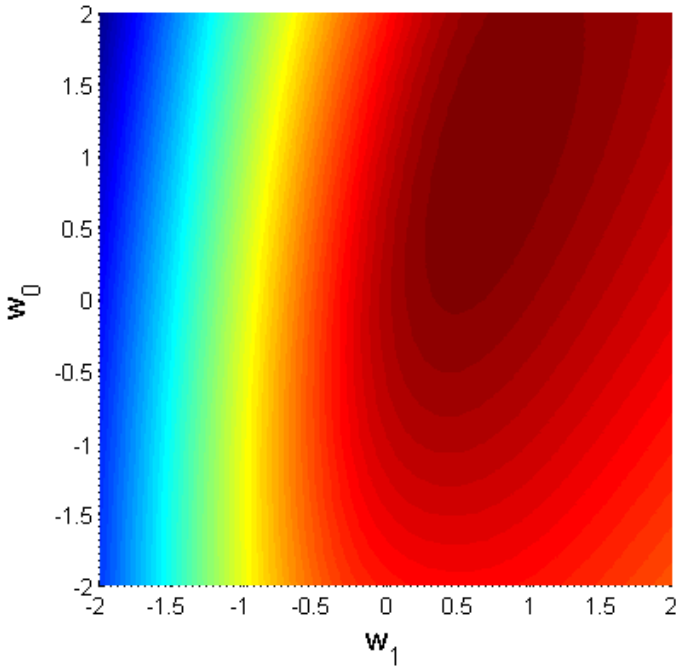


Classification dataset in 1-D. The vertical axis is the constant feature $X_0 = 1$.

The log likelihood of the data is

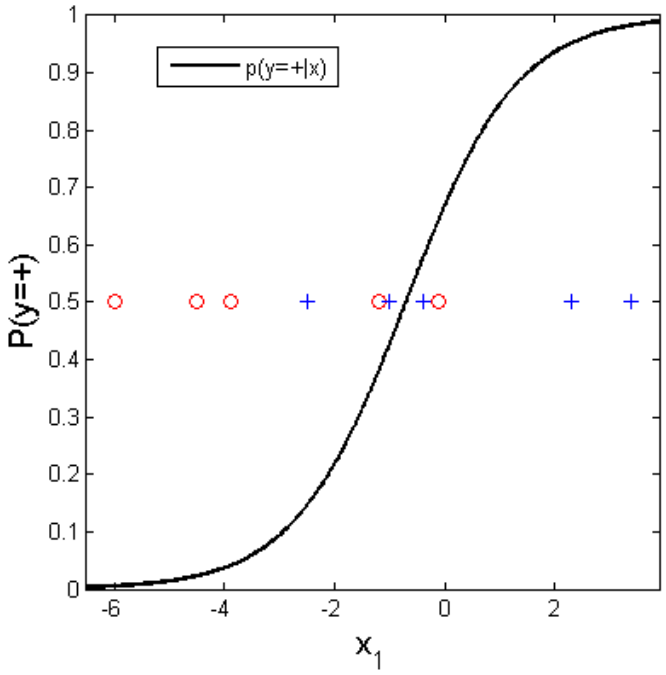
$$\ell(\mathbf{w}) = \log P(D_Y \mid D_X, \mathbf{w}) = \log\left(\prod_i \frac{1}{1+\exp\{-y_i \mathbf{w}^\top \mathbf{x}_i\}}\right) = -\sum_i \log(1+\exp\{-y_i \mathbf{w}^\top \mathbf{x}_i\}).$$

Shown below is the likelihood in the space of parameters w_0 and w_1, with maximum at $(w_0 = 1, w_1 = 0.7)$.



Log likelihood function heat map.

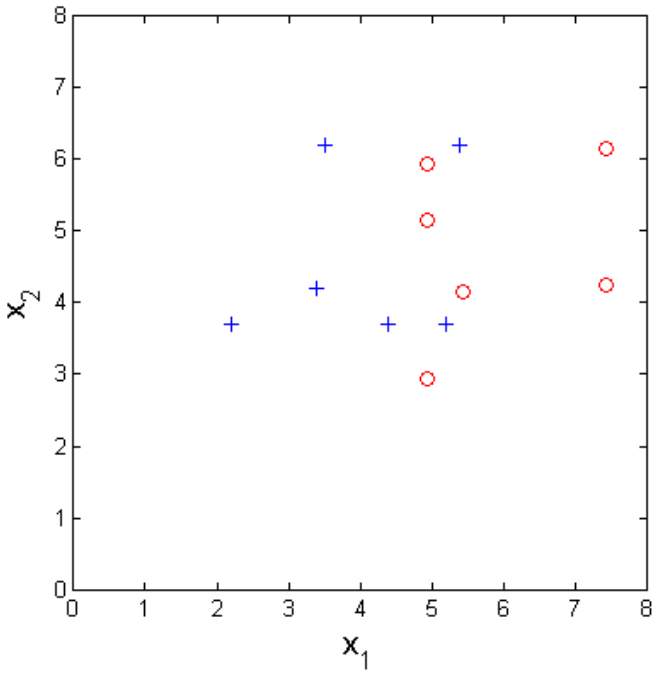
The MLE solution is displayed below.



Learned logistic regression.

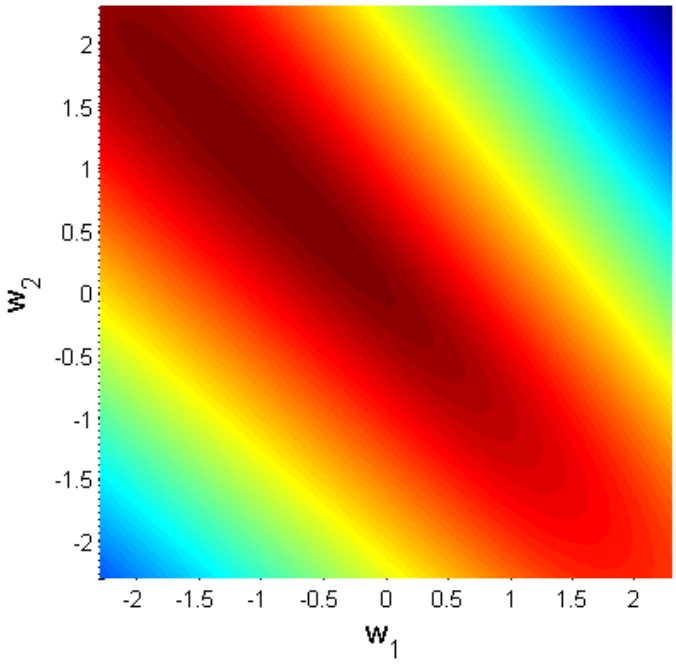
2 2-D Example

Now consider a 2-D problem, with positive examples marked as pluses and negative examples as circles. To visualize the problem in 2-D, we do not use the constant feature x_0.



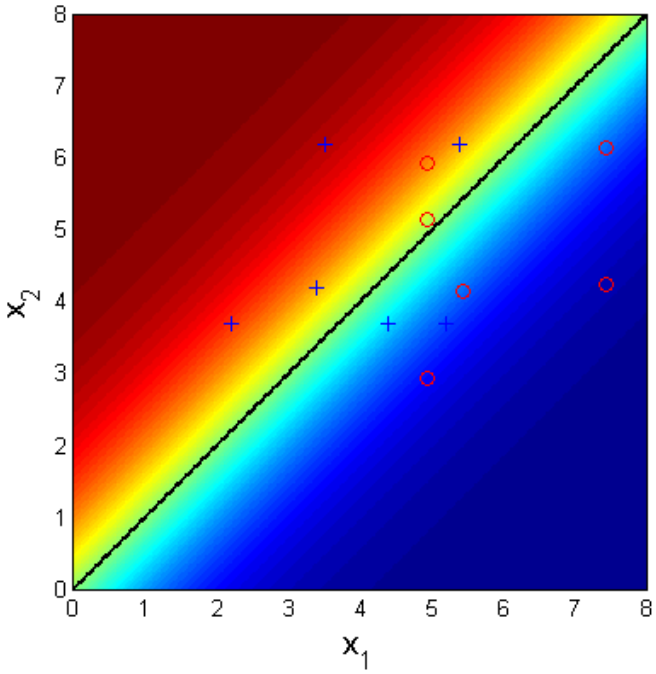
Classification dataset in 2 dimensions.

The log likelihood is the same as above. Its maximum in the space of w_1 and w_2 is $(-0.81, 0.81)$.



Log likelihood function heat map.

The MLE solution is shown below; red indicates high probability of the positive class, and the black line is the MLE decision boundary.



Learned logistic regression.

2.1 Linear Decision Boundary

Why is the boundary linear? At the boundary,

$$P(1 \mid X) = P(-1 \mid X) \quad \Longrightarrow \quad \mathbf{w}^\top \mathbf{X} = 0.$$

For the toy problem, the optimal $\mathbf{w}$ is $(-0.81, 0.81)$, so $-0.81x_1 + 0.81x_2 = 0$, yielding the line $x_1 = x_2$.

3 Computing MLE

There is no closed-form solution for the parameters as there is for Naive Bayes or (arguably) least-squares regression. To find the MLE we use iterative optimization. The objective is concave, so it has one global optimum (though it may be at infinity). Gradient ascent starts at, for example, $\mathbf{w}^0 = 0$ and climbs in the steepest direction. The gradient is

$$\nabla_{\mathbf{w}} \ell(\mathbf{w}) = \sum_i \frac{y_i \mathbf{x}_i \exp\{-y_i \mathbf{w}^\top \mathbf{x}_i\}}{1 + \exp\{-y_i \mathbf{w}^\top \mathbf{x}_i\}} = \sum_i y_i \mathbf{x}_i \left(1 - P(y_i \mid \mathbf{x}_i, \mathbf{w})\right).$$

Iterate the gradient-ascent update until the parameter change is below a chosen tolerance:

$$\mathbf{w}^{t+1} = \mathbf{w}^t + \eta \nabla_{\mathbf{w}} \ell(\mathbf{w}), \qquad \eta > 0.$$

4 Computing MAP

Assume an independent, zero-mean Gaussian prior with standard deviation λ for each parameter:

$$w_j \sim \mathcal{N}(0, \lambda^2), \qquad P(\mathbf{w}) = \prod_j \frac{1}{\lambda\sqrt{2\pi}} \exp\left\{-\frac{w_j^2}{2\lambda^2}\right\}.$$

Thus the MAP estimate maximizes

$$\log P(\mathbf{w} \mid D, \lambda) = \ell(\mathbf{w}) - \frac{1}{2\lambda^2} \mathbf{w}^\top \mathbf{w}.$$

The MAP gradient is

$$\nabla_{\mathbf{w}} \log P(\mathbf{w} \mid D, \lambda) = \sum_i y_i \mathbf{x}_i \left(1 - P(y_i \mid \mathbf{x}_i, \mathbf{w})\right) - \frac{1}{\lambda^2} \mathbf{w}.$$

The prior therefore penalizes large values of $\mathbf{w}$.

5 Multinomial Logistic Regression

For K classes, use $K-1$ sets of weights $\mathbf{w}_1, \ldots, \mathbf{w}_{K-1}$; the final set is redundant. For $k = 1, \ldots, K-1$,

$$P(Y = k \mid \mathbf{x}, \mathbf{w}) = \frac{\exp\{\mathbf{w}_k^\top \mathbf{x}\}}{1 + \sum_{k'=1}^{K-1} \exp\{\mathbf{w}_{k'}^\top \mathbf{x}\}}, \qquad P(Y = K \mid \mathbf{x}, \mathbf{w}) = \frac{1}{1 + \sum_{k'=1}^{K-1} \exp\{\mathbf{w}_{k'}^\top \mathbf{x}\}}.$$

With MAP estimation, a symmetric K-weight formulation is common:

$$P(Y = k \mid \mathbf{x}, \mathbf{w}) = \frac{\exp\{\mathbf{w}_k^\top \mathbf{x}\}}{\sum_{k'=1}^{K} \exp\{\mathbf{w}_{k'}^\top \mathbf{x}\}}, \quad k = 1, \ldots, K.$$

6 Naive Bayes vs. Logistic Regression

	Naive Bayes	Logistic Regression
Assumption	$P(\mathbf{X} \mid Y)$ is simple	$P(Y \mid \mathbf{X})$ is simple
Likelihood	Joint	Conditional
Objective	$\sum_i \log P(y_i, \mathbf{x}_i)$	$\sum_i \log P(y_i \mid \mathbf{x}_i)$
Estimation	Closed form	Iterative (gradient, etc.)
Decision boundary	Quadratic/linear	Linear
When to use	Very little data vs. parameters	Enough data vs. parameters

7 Linear Boundary for 2-Class Gaussian Naive Bayes with Shared Variances

Gaussian Naive Bayes typically estimates a separate variance σ_{jk} for every feature j and class k. If variances are shared, there is one parameter σ_j per feature. The class distributions then have the same density-contour shape, and the decision boundary is linear:

$$P(Y = 1 \mid X) = \frac{1}{1 + \exp\{w_0 + \mathbf{w}^\top X\}}.$$

Let $P(Y = 1) = \theta$ and

$$P(X_j \mid Y = 1) = \frac{1}{\sigma_j \sqrt{2\pi}} \exp\left\{ -\frac{(X_j - \mu_{j1})^2}{2\sigma_j^2} \right\}, \quad P(X_j \mid Y = 0) = \frac{1}{\sigma_j \sqrt{2\pi}} \exp\left\{ -\frac{(X_j - \mu_{j0})^2}{2\sigma_j^2} \right\}.$$

Then

$$\begin{aligned} \log \frac{P(Y = 0)P(X \mid Y = 0)}{P(Y = 1)P(X \mid Y = 1)} &= \log \frac{1 - \theta}{\theta} + \sum_j \frac{(\mu_{j0} - \mu_{j1})X_j}{\sigma_j^2} + \frac{\mu_{j1}^2 - \mu_{j0}^2}{2\sigma_j^2} \\ &= w_0 + \mathbf{w}^\top X. \end{aligned}$$

Here capital X and X_j denote random variables, not matrices.